**T6** $\xi \sim p(x) = \frac{\theta-1}{x^\theta}\{x \geq 1\}$, $\theta > 1$

$\vec{x}_n$ - выборка

а) $L(\theta) = \prod_{i=1}^{n} p(x_i, \theta) = \frac{(\theta-1)^n}{(\prod_{i=1}^{n} x_i)^\theta}\{x_{(1)} \geq 1\}$

$(\ln L)' = (n\ln(\theta-1) - \theta\sum_{i=1}^{n}\ln x_i)' =$

$= \frac{n}{\theta-1} - \sum_{i=1}^{n}\ln x_i = 0 \Rightarrow \tilde{\theta} = 1 + n(\sum_{i=1}^{n}\ln x_i)^{-1}$

$\Rightarrow \tilde{\theta} = 1 + (\overline{\ln x})^{-1}$, где $\overline{\ln x} = \frac{1}{n}(\sum_{i=1}^{n}\ln x_i)$

$(\ln \theta)'' = -\frac{n}{(\theta-1)^2} < 0 \Rightarrow$ максимум

Итак, $\tilde{\theta} = 1 + (\overline{\ln x})^{-1}$ - ОМП

б) Исследуем вероятностную модель на сильную регулярность

1) • $p(x, \theta)$ непрер. дифф. по $\theta$ на $\theta > 1$

$\frac{\partial p}{\partial \theta} = \frac{1 - \ln x(\theta-1)}{x^\theta}$

$\Rightarrow \int_{-\infty}^{\infty}\frac{\partial p}{\partial \theta}dx = \int_{-\infty}^{\infty} x^{-\theta}(1 - \ln x(\theta-1))dx =$

$= \int_{-\infty}^{\infty} x^{-\theta}dx - \int_{-\infty}^{\infty} x^{-\theta}\ln x(\theta-1)dx =$

$= \frac{1}{1-\theta}x^{-\theta+1}\Big|_{-\infty}^{\infty} - (\theta-1)\int_1^{\infty} x^{-\theta}\ln x\,dx =$

$\frac{1}{1-\theta} - (\theta-1)\cdot\frac{1}{(\theta-1)^2} = 0 \Rightarrow \int_{-\infty}^{\infty}\frac{\partial p}{\partial \theta}dx = \frac{\partial}{\partial \theta}\int_{-\infty}^{\infty} p(x)dx$

• $\ln p = \ln(\theta-1) - \theta\ln x$

$\frac{\partial \ln(p)}{\partial \theta} = \frac{1}{\theta-1} - \ln x$

$I(\theta) = \int_1^{+\infty}\frac{\theta-1}{x^\theta}(\frac{1}{\theta-1} - \ln x)^2 dx = \int_1^{+\infty}\frac{\theta-1}{x^\theta}(\frac{1}{(\theta-1)^2} - \frac{2\ln x}{\theta-1} + \ln^2 x)dx$

$$= \int_1^{+\infty} \frac{dx}{x^\theta(\theta-1)} - 2\int_1^{+\infty} \frac{\ln x}{x^\theta}dx + \int_1^{+\infty} \frac{(\theta-1)\ln^2 x}{x^\theta}dx$$

$$I(\theta) = \int_1^{+\infty} \frac{\theta-1}{x^\theta}\left(\frac{1}{\theta-1} - \ln x\right)^2 dx = \left\langle \begin{matrix} y = \ln x, & x = e^{xy} \\ dy = \frac{dx}{x} & dx = e^y dy \end{matrix} \right\rangle$$

$$= \int_0^{+\infty} (\theta-1)e^{-\theta y}\left(\frac{1}{\theta-1} - y\right)^2 e^y dy =$$

$$= \int_0^{+\infty} (\theta-1)e^{y(1-\theta)}\left(\frac{1}{\theta-1} - y\right)^2 dy. \quad \Big] \ \lambda = \theta - 1, \text{ тогда}$$

$$I(\theta) = \int_0^{+\infty} \lambda e^{-\lambda y}\left(\frac{1}{\lambda} - y\right)^2 dy = \int_0^{+\infty} \frac{e^{-\lambda y}}{\lambda}dy - 2\int_0^{+\infty} e^{-\lambda y} y\, dy +$$

$$+ \lambda \int_0^{+\infty} e^{-\lambda y} y^2 dy = \frac{1}{\lambda^2} - 2\frac{1}{\lambda^2} + \frac{2}{\lambda^2} = \frac{1}{\lambda^2} = \frac{1}{(\theta-1)^2} -$$

непрер. и $>0$ на $\theta > 1$

$\Rightarrow$ модель регулярна

2) $p(x,\theta)$ – $n \geq 2$ раз непрер. дифф. по $\theta$ на $\theta > 1$

3) $\frac{\partial^l p}{\partial \theta^l}$ (для $l=2$): $\ln x \cdot x^{-\theta}(\theta \ln x - \ln x - 2)$

$$\int_1^{+\infty} \frac{\partial^2 p}{\partial \theta^2} dx = \int_1^{+\infty} (\ln^2 x \cdot x^{-\theta}(\theta-1) - 2\ln x \cdot x^{-\theta})dx = 0 -$$

выполнено $\forall l$

$\Rightarrow$ ОМП состоят., асимп. несмещ., асимп. эффективна и асимп. нормальна

Медиана по определению: $F(x_{0,5\frac{1}{2}}) = \frac{1}{2}$

$$\int_1^{x_{\frac{1}{2}}} \frac{\theta-1}{x^\theta}dx = -\frac{1}{x_{\frac{1}{2}}^{\theta-1}} + 1 = \frac{1}{2} \Rightarrow x_{\frac{1}{2}} = 2^{\frac{1}{\theta-1}}$$

Тогда $g(\tilde\theta) = \sqrt[\tilde\theta-1]{2}^{\,x_{\frac{1}{2}}} \Rightarrow \sqrt{n}\,\frac{g(\hat\theta) - g(\theta)}{\sqrt{\nabla^T g(\tilde\theta)\, I^{-1}(\theta)\, \nabla g(\tilde\theta)}} \rightsquigarrow N(0,1)$

$$\nabla g(\tilde\theta) = 2^{\frac{1}{\tilde\theta-1}} \ln 2 \cdot \left(-\frac{1}{(\tilde\theta-1)^2}\right)$$

$$\sqrt{n}\,\frac{(g(\tilde\theta)-g(\theta))(\tilde\theta-1)}{2^{\frac{1}{\tilde\theta-1}}\ln 2} \sim N(0,1)$$

$$\Rightarrow P\left(g(\tilde\theta)-\frac{1{,}96\ln 2\cdot 2^{\frac{1}{\tilde\theta-1}}}{\sqrt{n}(\tilde\theta-1)} < x_{\frac{1}{2}} < g(\tilde\theta)+\frac{1{,}96\ln 2\cdot 2^{\frac{1}{\tilde\theta-1}}}{\sqrt{n}(\tilde\theta-1)}\right) =$$

$= \beta$ – асимптот. довер. интервал для медианы

б) $$\frac{\sqrt{n}(\tilde\theta-\theta)}{\tilde\theta-1} \sim N(0,1)$$

$$-1{,}96 < \sqrt{n}\,\overline{\ln x}\,(1+(\overline{\ln x})^{-1}-\theta) < 1{,}96$$

$$P\left(1-\frac{1{,}96-\sqrt{n}}{\sqrt{n}\,\overline{\ln x}} < \theta < 1+\frac{1{,}96+\sqrt{n}}{\sqrt{n}\,\overline{\ln x}}\right) = \beta -$$

асимп. довер. интервал для $\theta$